

THE UNIVERSITY OF
SYDNEY

Tutorial Class	
Student Name	
SID	

CONFIDENTIAL QUIZ PAPER

This paper should not be removed from the quiz venue.

COMP5339 – Data Engineering

Tutorial Quiz – 1

Individual Quiz (5%)

Quiz Writing Time: **30 minutes**
Friday, 17 April, at 07 PM

Instructions to Students

- Please ensure you correctly complete your **tutorial** and **Student ID** details.
- This is a **closed-book** test. **Electronic devices** (including phones, laptops, tablets, etc.) are **not permitted**. Please place all devices in your bag and keep your bag on the floor.
- This tutorial quiz consists of **3 multiple-choice questions** and three short descriptive questions. **All questions must be answered.**
- Write all answers in the spaces provided on this paper.
- Questions carry **different marks**. The mark value for each question is shown at the start of the question. The **total mark** for this quiz is **10**, which contributes to 5% of the total marks.
- For short-answer questions, please write **legibly**. Record your final answers in **ink**. **Do not** use a pencil or **red ink**.

Please submit the completed quiz paper to your tutor **before leaving the room**.

Part I – Multiple Choice Questions

Question 1: This question has three parts. Tick the boxes corresponding to each correct answer. Each question carries one mark.

A. In a data pipeline, what is the main challenge when extracting data from multiple linked webpages?

- ☐ Data storage
- ☐ Network speed
- ☐ **Maintaining consistent structure across pages**
- ☐ Visualising the data

B. In a document database (such as MongoDB), relationships between entities can be represented using embedding or referencing. When is embedding preferred?

- ☐ When data is highly normalised
- ☐ **When related data is frequently accessed together**
- ☐ When relationships are many-to-many and large
- ☐ When strict ACID transactions are required

C. Why are APIs generally preferred over web scraping in data engineering?

- ☐ **APIs provide structured and stable access to data**
- ☐ APIs provide unstructured data
- ☐ APIs are always free
- ☐ APIs require more manual effort

Part II – Short Answer Questions

Scenario 1: You are given a webpage from an online unit allocation, presented in an HTML structure below. You need to extract this data for a data warehouse.

Answer the following:

1. Identify the attributes and write at least one record (1 Mark)
2. Which elements will you target for extraction? (1 Mark)

```
<!DOCTYPE html>
<html>
<head>
<title>Unit Allocations</title>
</head>
<body>

<div id="courses">
  <div class="course">
    <h2 class="title">Data Engineering</h2>
    <span class="code">COMP5339</span>
    <span class="lecturer">Imdad</span>
    <span class="enrolments">450</span>
  </div>

  <div class="course">
    <h2 class="title">Machine Learning</h2>
    <span class="code">COMP5310</span>
    <span class="lecturer">Lee</span>
    <span class="enrolments">200</span>
  </div>

  <div class="course">
    <h2 class="title">Databases</h2>
    <span class="code">COMP9123</span>
    <span class="lecturer">Imdad</span>
    <span class="enrolments">870</span>
  </div>
</div>

</body>
</html>
```

Attributes: title, code, lecturer, enrolments

Record: Data Engineering, COMP5339, Imdad, 450

Element to target for extraction: div id = "course"

Question 2: Suppose the webpage changes to (compared to the initial outline, above) the following:

```
<div class="course">
  <h2 class="title">Data Engineering</h2>
  <span class="code">COMP5339</span>
  <span class="lecturer">Imdad </span>
  <span class="enrolments">450 students</span>
</div>
```

What data quality issues may arise, and how would you handle them in a pipeline? (2 Marks)

Issues:

- The numeric field under the "enrolments" becomes text
- Inconsistent format

How to handle them:

- Apply transformation to extract numeric values
- Standardise format (e.g., remove “students”) and enforce schema validation

Question 3: Suppose the webpage structure changes to (compared to the initial structure, above):

```
<div class="course-item">
  <h2 class="title">Data Engineering</h2>
  <span class="code">COMP5339</span>
  <span class="lecturer">Imdad </span>
  <span class="enrolments">450</span>
</div>
```

1. What has changed? (1 Mark)
2. What problem does this cause, and how would you make your pipeline more robust? (1 Mark)

What has changed: class = "course" has now changed to class = "course-item"

Part 2 - Sample answer:

Problem:

- Extraction logic breaks due to dependency on class name

Solutions:

- Add validation checks
- Implement monitoring/alerts
- Design adaptable extraction rules

Question 4: The website owner later provides an API that returns structured course data. Will you switch from scraping to API? Justify your answer. **(1 Mark)**